

BUILD PROMPT — Hunting Map Generator (Desktop App)

How to use this file: Paste this entire document as your first message to Claude Code (running Claude Fable). Everything the model needs is here. Work through it in the order given in Section 14.

0. READ THIS FIRST — Non-Negotiable Ground Rules

1. **The end user is not a developer.** After they download the app, they must never open a terminal, never run a command, never install Node, Python, Rust, GDAL, or any other dependency. They double-click an installer, then double-click an icon. That's it. If any feature would require the user to touch a CLI, redesign that feature.
 2. **All dependencies must be bundled inside the installer.** No post-install setup steps, no "first run, please install X" prompts.
 3. **Everything is local. There is no cloud.** No accounts, no logins, no sign-up, no sync, no telemetry, no analytics, no crash reporting to a server, no server-side component of any kind. The only network traffic the app ever makes is fetching public map data the user explicitly asked for. Fully functional offline once data is downloaded.
 4. **Zero paid services.** No Mapbox, no Regrid, no ATTOM, no API keys, no usage-metered tile providers. Public/government data only. If a layer is only available from a paid source, drop it and note it in the README.
 5. **This is a map *printing* tool, not a navigation app.** It generates static, print-ready PDFs. It is not used in the field.
 6. **The app classifies land access, it does not give legal advice.** See Section 6 — the most safety-critical part of the build, with its own hard rules.
 7. **Build iteratively.** Complete each stage in Section 14, confirm it works, then move on. Do not write the whole app in one pass.
 8. **Ask before assuming.** If a data source is ambiguous or a requirement here is unclear, stop and ask rather than guessing.
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1. The Workflow — Build Toward This Exactly

This is the canonical user journey. The entire interface exists to serve it, in this order.

1. **New Project** — from the Projects home screen.
2. **Format** — pick paper size and orientation, or choose to draw a freeform area instead.
3. **Number of maps** — choose how many sheets cover the area: one map, two maps, a 3×2 grid, or custom. This is an *input*, not something the app decides for the user.
4. **Layers** — choose what appears on the map, so the terrain is visible while positioning.
5. **Place the box** — zoom to the hunting area. A crisp rectangle sits on the map showing exactly what will print. **Everything outside it is visibly dimmed.** The paper margin band is distinctly shaded so it's obvious where map content stops and the legend/title area begins. If the area spans multiple sheets, **division lines show where the page splits fall**, with the overlap zone marked.
6. **Preview** — page-by-page preview of the finished map, with **checkboxes right there** to toggle grid lines, coordinate labels, compass rose, scale bar, legend, title block, and contour labels. Changes appear in the preview immediately.
7. **Download** — export the PDF.

Separately, and reachable without creating a project at all: a **Data screen** for downloading county map data ahead of time.

2. Platforms & Distribution

Must build and run natively on:

- **Windows 10/11** (x64) — `.msi` or `.exe` installer
- **macOS** (Apple Silicon and Intel) — `.dmg`
- **Ubuntu Linux 22.04+** — both `.deb` and `.AppImage`

Requirements:

- Installers produced automatically by the build pipeline.
- Proper app icon; appears in the OS app launcher / Start menu / Applications folder.
- `BUILDING.md` for the project owner (how to produce installers, plus what macOS notarization and Windows code-signing would involve — not required for working builds)

and a plain-English `README.md` for end users.

3. Tech Stack

- **App shell:** Tauri v2 (preferred). Use Electron only if Tauri presents a blocking issue — tell me if you switch and why.
 - **Frontend:** React + TypeScript.
 - **Map rendering:** MapLibre GL JS (open source, no API key, no usage billing).
 - **Local storage:** SQLite (Tauri SQL plugin) for the catalog, download manifest, and settings index. Projects and downloaded data on disk per Section 4.
 - **PDF generation:** `pdf-lib` or equivalent. Render map imagery at target DPI; compose cartographic elements as vector overlays so text and lines stay crisp.
 - **Data formats consumed:** XYZ/WMTS raster tiles, GeoJSON, Shapefile (convert to GeoJSON on import — bundle the conversion, never make the user do it).
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4. Local Storage Layout

All data in the OS-appropriate application data directory, with a setting to relocate it. Human-readable and portable — backing up means copying a folder.

```
HuntingMaps/  
  projects/  
    <project-id>/  
      project.json          human-readable project definition  
      thumbnail.png         auto-generated preview for the Projects screen  
      snapshots/            rolling undo history  
      exports/              PDFs exported from this project  
  data/  
    colorado/  
      _state.json           county index and metadata  
      larimer/  
        *.mbtiles           raster tiles  
        *.geojson           vector layers  
        _manifest.json      source dates, sizes, checksums  
    wyoming/  
      ...  
  settings/  
    settings.json
```

presets/	user-saved color and print presets
regions/	layer configs (Section 10)
access-rules/	classification rules (Section 6)
logs/	

- `project.json` must be readable, hand-editable JSON — no opaque binary formats.
 - Deleting a project folder removes it cleanly.
 - Provide in-app **Backup** (zip the folder to a chosen location) and **Restore**, so no one navigates the filesystem.
 - The app never writes outside this directory and the user's chosen export locations.
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5. Map Projects

5.1 Projects screen (app home)

A grid of project cards, each showing: auto-generated thumbnail, project name, state and counties covered, area size, last modified, sheet count (e.g. "6 sheets"), and a **fork lineage badge** if applicable ("Forked from *Poudre Canyon 2025*").

Controls: **New Project**, **Download County Data** (jumps straight to the Data screen — Section 9), search box, sort by name/created/modified, optional folders or tags. Per-card menu: **Open** · **Fork** · **Rename** · **Export Project File** · **Delete**.

5.2 Forking

Forking creates an independent copy so the original is never disturbed — for trying a different scale, color scheme, or shifted area without losing what worked.

- Fork from the project card menu or from inside an open project.
- The fork gets a new ID and a name defaulting to `<original> (fork)`, editable at creation. It copies area, format, layers, print settings, and element toggles.
- `project.json` stores a `forkedFrom` reference (parent ID and name). Show lineage both ways: a fork displays "Forked from X"; a parent lists its forks. Both are clickable.
- Deleting a parent must not break its forks — they become standalone with the note "Forked from a deleted project."
- **Forking copies project settings only, not downloaded map data** — data is shared from the common `data/` store, so a fork costs almost no disk space. Say so in the UI so

users fork freely.

5.3 Project export/import

- **Export Project File** writes a single portable `.huntmap` file (zipped `project.json` + thumbnail) that can be emailed or copied to another machine.
- **Import** reads it back; if the machine lacks required county data, prompt to download it (Section 9.3).
- This is the only sharing mechanism — file-based, no server.

5.4 Autosave and snapshots

- Projects autosave continuously. No Save button to forget.
- Keep a rolling **snapshot history** (default last 20 states) with an **Undo changes / Restore snapshot** panel, so an experiment can be walked back without forking. Snapshots are local, in the project folder, retention configurable in Settings.

6. Land Access Classification — Safety-Critical

The feature that makes the app genuinely useful, and the one most capable of causing real harm if done carelessly. Read this whole section before writing any of it.

6.1 The five access categories

Every land polygon is classified into exactly one:

#	Category	Meaning in plain English	Default color
1	Open — Hunting Allowed	Public land you can enter and hunt on, subject to season and license	Green
2	Open — No Hunting	Public land you can walk on, but hunting is prohibited	Blue
3	Closed — No Public Entry	Public or restricted land you may not enter at all	Red
4	Private Land	Private property — permission required	Orange/tan
5	Unknown / Unclassified	Access status could not be determined from available data	Grey, diagonal hatching

Plus a separate line layer, not a polygon fill:

Layer	Meaning	Default style
Public Access Routes	Roads, trails, and easements the public may legally travel, including routes crossing private land	Purple dashed line

6.2 How classification is derived — and its limits

Critical: source datasets contain no “can I hunt here” field. Classification comes from a **rules engine** mapping managing agency, designation, and access attributes onto the five categories.

- Store rules in editable, human-readable JSON: `access-rules/_national.json`, `access-rules/colorado.json`, `access-rules/wyoming.json`. Each rule maps source attribute values to a category with a plain-English note explaining the reasoning.
- Rules evaluate in priority order; state rules override national defaults.
- **Anything the rules cannot confidently classify becomes Category 5 (Unknown), never a guess.** Never default unmatched polygons to “Open.” Ambiguity fails toward caution.
- Every polygon retains a hidden **source trace** — which dataset and attribute produced its classification — surfaced in a click-to-inspect popup so the user can see *why* something is colored as it is.

Starting rules (expand and document each):

- BLM and USFS general land → Category 1
- National Park Service units → Category 2, unless a hunt-permitted unit (then Category 1)
- National Wildlife Refuges → Category 2 by default; Category 1 only where explicitly hunt-open in source data; **never assume**
- Military reservations, restricted federal facilities → Category 3
- Colorado State Trust Land → **Category 3 by default**, upgraded to Category 1 only where the parcel appears in CPW’s public-access lease dataset
- Wyoming State Trust Land → Category 1 where state law allows general access, verified against current WGFD data and documented in the rule note
- CPW State Wildlife Areas → Category 1, but SWAs with special restrictions flagged

Unknown rather than assumed

- Walk-In Access / Hunter Management Areas → Category 1, noted as seasonal
- Parcels with no public land designation → Category 4
- Everything else → Category 5

6.3 Mandatory user-facing warnings

Not optional, not user-disableable:

1. **On every exported PDF page**, a margin disclaimer block reading approximately:

Land access shown is approximate and derived from public datasets. It is for planning only and is not legal authority. Boundaries may be inaccurate and access status changes. Verify current regulations, closures, and land status with CPW / WGFD and the managing agency before hunting. Data current as of [date]. Map generated [date].

2. **A first-run modal** explaining the same, acknowledged once.
3. **A persistent note** at the bottom of the layer panel whenever any access layer is visible:
"Access categories are estimates — verify before you hunt."
4. **Data currency stamps.** Store and display each dataset's source date. Flag anything older than 12 months as "May be outdated." Print the data date on the PDF.
5. **The legend must always include the Unknown category** whenever unclassified polygons fall inside the printed area.

6.4 Known limitations the app must surface

In a plain-English "**What this map can't tell you**" help page, linked from the project workspace and referenced in the printed disclaimer:

- Public land is not automatically huntable — closures, safety zones near roads and buildings, and seasonal restrictions are not shown
- Landlocked public parcels with no legal access route are not identified
- Checkerboard land patterns (common in Wyoming) create corner-crossing questions this app does not answer
- Parcel boundaries are frequently inaccurate by tens of feet or more; they are not surveys
- Ownership and boundaries change; a printed map ages

- Walk-in and access-program properties are seasonal, and enrollment changes annually

7. Map Layers

Defined in config files, **never hardcoded into UI or map logic**. See Section 10.

7.1 National layers (available everywhere)

Layer	Source	Type
USGS Topo basemap	USGS National Map (USGSTopo tile service)	Raster tiles
USGS Imagery (aerial)	USGS National Map imagery service	Raster tiles
NAIP aerial imagery	USDA NAIP via USGS/USDA public service	Raster tiles
Shaded relief / hillshade	USGS 3DEP-derived hillshade	Raster tiles
Contour lines	USGS 3DEP contours, or generated from 3DEP DEM	Vector
Slope shading	Derived from 3DEP DEM	Raster/derived
Land management boundaries	PAD-US (USGS Protected Areas Database)	Vector
Roads	USGS National Map transportation / OpenStreetMap	Vector
Trails	USGS National Map / OpenStreetMap	Vector
Motor vehicle routes (MVUM)	USFS Motor Vehicle Use Map data	Vector
Water features	USGS National Hydrography Dataset (NHD)	Vector

7.2 Region-specific layers

Colorado (`regions/colorado.json`)

- Parcel boundary lines — Colorado statewide parcel data via Colorado Information Marketplace / DOLA GIS clearinghouse. **Boundaries only.**
- GMU boundaries — Colorado Parks & Wildlife open data portal

- State Wildlife Areas — CPW
- State Trust Land, including CPW public-access-leased parcels — Colorado State Land Board / CPW
- Walk-In Access / Public Access Program properties — CPW

Wyoming (`regions/wyoming.json`)

- Parcel boundary lines — county assessor GIS data via Wyoming Geographic Information Science Center (WyGIS). **Boundaries only.**
- Hunt Area boundaries — Wyoming Game & Fish Department
- State Trust Land — Wyoming Office of State Lands and Investments
- Walk-In Areas and Hunter Management Areas — WGFD
- Public access easements — WGFD / county data where available

7.3 Hard rule on ownership data

Import **boundary geometry only**. On import, actively strip owner names, mailing addresses, tax IDs, and all other ownership attributes. Never display or store them. The map shows *where a boundary is*, not *who owns it*.

8. Project Workspace — Tabs

Opening a project fills the window with a tabbed workspace. **Tab order mirrors the workflow in Section 1.** A header above the tabs shows the project name (click to rename), state and counties, online/offline toggle, autosave indicator, and a **Fork** button.

Every tab has its own settings. A gear icon sits in the top-right corner of each tab's panel, opening a settings drawer scoped to that tab and to this project only. See Section 12.2 for how these relate to global settings.

Tabs remember scroll position and state when switching. Unfinished prerequisites show a subtle dot on the tab (Preview is dotted until an area exists) — a hint, never a hard block. Keyboard shortcuts Ctrl/Cmd+1 through 6 switch tabs.

Tab 1 — Format

Decide the shape of the output before touching the map.

Paper

- Size: Letter (8.5×11), Legal, Tabloid (11×17), A4, A3, plus **Custom** (user enters dimensions)
- Orientation: Portrait / Landscape
- Margins: Narrow / Normal / Wide, with a small diagram showing the effect

Sheet layout — “How many maps?” This is a direct choice, presented as large clickable option cards, not a computed afterthought:

- **1 sheet** — the whole area on one page
- **2 sheets** — with a Side-by-side / Stacked choice
- **4 sheets** (2×2)
- **6 sheets** (3×2 or 2×3)
- **Custom grid** — user enters columns × rows
- **Auto** — let the scale decide how many sheets are needed

Scale

- Standard ratios (1:12,000 · 1:24,000 · 1:31,680 · 1:50,000 · 1:63,360 · 1:100,000) plus Custom, each labeled in plain English, e.g. *“1:24,000 — about 2.6 inches per mile”*
- A **Lock** toggle. This governs how Format and Area interact, and must be explained inline in one sentence each:
 - **Scale locked** — the box on the map is a fixed ground size. Making the area bigger adds sheets.
 - **Sheet count locked** — the box resizes freely and the scale adjusts to fit. Dragging a bigger area gives a smaller scale.
- Whichever is locked, the current values for scale, sheet count, and ground coverage are displayed together and update live.

Freeform option A **“Draw my own area instead”** toggle. When on, the user drags any rectangle and the app reports the resulting scale and sheet count afterward rather than constraining the box up front.

Tab settings (gear): default paper size, default orientation, default sheet layout, default scale, custom paper presets, whether Auto or a fixed sheet count is the default for new projects.

Tab 2 — Layers

Choose what appears on the map, before positioning, so the terrain is visible while placing the box.

- Checkbox list grouped under headers: **Basemap · Terrain · Land Access · Features · Boundaries**
- Basemap options are radio buttons (one at a time); everything else is a checkbox
- An opacity slider appears under a checked layer
- The **Land Access** group shows each of the five categories with its color swatch inline and a gear icon linking to access color settings
- Per-project style overrides live here — a project can differ from the global default without changing it
- Clicking any polygon on the map opens the access-category popup with its source trace (6.2)
- Persistent note at the panel bottom: *“Access categories are estimates — verify before you hunt.”*

A compact version of this list is also reachable from the Area tab, so layers can be toggled while positioning without switching tabs.

Tab settings (gear): default layers on for new projects, default opacities, label sizes, access color preset for this project, whether the access popup opens on click.

Tab 3 — Area

The centerpiece. Zoom to the hunting area and position the print box.

The box and its shading — build this precisely:

Three visually distinct zones on the map:

1. **Inside the print extent** — full brightness, unobstructed. This is what will appear on the map.
2. **The paper margin band** — a distinct band just inside the box edge, lightly shaded or hatched, showing where the paper margin sits and map content will not reach. Legend, title block, and disclaimer live here.

3. **Everything outside the box** — visibly dimmed with a dark translucent overlay (default 50%, adjustable in tab settings), so the print area reads instantly at a glance.

Draw a crisp **neatline** at the print boundary. The box has draggable corner and edge handles.

Division lines for multi-sheet maps:

- Dashed lines across the box showing exactly where sheet splits fall
- The **overlap zone** between sheets shown as a narrow shaded band, so the user sees the duplicated strip
- Each cell labeled with its sheet reference (A1, A2, B1...) drawn on the map
- Live readout: **"6 sheets (3 across × 2 down) at 1:24,000 on Tabloid"**

Four ways to define the area — all available, switchable without losing work:

1. **Position the format box** — the rectangle from the Format tab appears sized to the chosen paper, scale, and sheet layout. Drag to position. Dragging a corner rescales it (or resizes the area, depending on which lock is set), with scale and sheet count updating live.
2. **Draw a box** — freeform drag, editable afterward.
3. **Place 4 corners** — click four points to set corners individually, each with a numbered draggable handle. If the corners imply rotation, offer **Snap to North-Up** or **Keep Rotated**. Live lat/long shown per corner.
4. **Enter coordinates** — numeric fields for N/S/E/W bounds or four corner pairs, accepting decimal degrees and degrees-minutes-seconds. Typing moves the box; moving the box updates the fields. Always two-way synced.

Also on this tab: search bar accepting place names and coordinates, live area size readout, counties covered with download status, and the missing-county banner from 9.3.

Tab settings (gear): outside-area dim opacity, margin band style (shaded/hatched/outline), division line color and weight, whether sheet labels draw on the map, whether the overlap band is shown, snap-to-north default, coordinate entry format (decimal vs DMS).

Tab 4 — Preview

Page-by-page preview of the finished map, exactly as it will export. **Element checkboxes live here**, in a panel beside the preview, so their effect is visible immediately.

Preview area:

- Thumbnail strip of all sheets; click or arrow-key to move between them
- Large view of the selected sheet, zoomable
- What you see is precisely what exports — no surprises

Element checkboxes (all on by default, each toggles live):

- Grid lines — with a picker for None / UTM / MGRS / Latitude-Longitude
- Coordinate labels at page edges
- Compass rose (true and magnetic north, with declination in degrees)
- Scale bar
- Ratio scale text
- Legend
- Title block
- Contour lines
- Contour elevation labels
- Sheet reference and page number
- Index inset (small map showing which sheet this is)
- Edge match markers (which sheet adjoins each side)
- Overview sheet as first page
- Neatline
- Notes printed on the overview sheet
- **Access disclaimer block — shown, checked, and locked. Cannot be turned off** (see 6.3)

Tab settings (gear): compass rose style, scale bar units and length, grid line color and weight, legend position and box style, title block contents, contour label size, neatline weight, preview render quality.

Tab 5 — Export

- Output: one multi-page PDF, or one PDF per sheet in a folder
- DPI: 150 / 200 / 300 / 400, noting higher means slower and larger
- Output folder picker, remembered between exports
- Filename pattern with tokens (project name, date, sheet reference)
- **Export** button, progress bar, and a **Cancel** button — high-DPI multi-sheet sets take time and the app must never look frozen
- On success: confirmation with **Show in Folder** opening the OS file browser at that location
- Export history for this project, each entry with its own Show in Folder
- Embed georeferencing metadata in the PDF where feasible (nice-to-have, not a blocker)

Tab settings (gear): default DPI, default output folder, default filename pattern, single-file vs per-sheet default, whether to open the folder automatically after export.

Tab 6 — Notes

Plain text notes saved with the project — why this area, what changed between forks, what to check next season. Optionally printed on the overview sheet.

Tab settings (gear): print notes on overview sheet by default, note font size when printed.

9. The Data Screen

Reachable from the Projects home screen and the main navigation, **independent of any project**. Downloading county data must never require creating a map.

9.1 County selection — map and list, side by side

Two synchronized views. A toggle switches Map / List / Both.

Map view (primary):

- Interactive map of the selected state with **county polygons drawn and labeled**
- Click a county to select or deselect
- Lasso drag or shift-drag box to select several

- **Status colors**, with a legend:
 - Downloaded and current → solid green
 - Downloaded but past the currency threshold → amber
 - Selected, not yet downloaded → blue outline
 - Not downloaded, not selected → light grey
 - Partially downloaded → green with hatching
- Hover tooltip: county name, status, estimated size, source date
- Buttons: **Select All** · **Clear** · **Select Downloaded** · **Invert**

List view:

- Alphabetical checkbox list with matching status colors, size, and source date
- Filter box
- Hovering a row highlights that county on the map and vice versa — always in sync

9.2 Layer selection and downloading

- Layer checkboxes with per-layer size estimates
- Running total: *“Selected: 3 counties, 6 layers — approximately 1.4 GB”*
- **Download** button with per-item and overall progress
- Downloads **pause, resume, and survive a dropped connection** without corrupting existing data
- **Manage Downloads**: what’s on disk, source date, size, per-item **Update** and **Delete**, “May be outdated” flags
- Total disk usage and storage location with **Change Location**

9.3 Missing-county prompt (required behavior)

Whenever a project’s area is set or changed, compute which counties it intersects and compare against what’s downloaded.

If anything is missing, show a **non-blocking banner** inside the project — never an interrupting modal:

Your map area covers Larimer County, which isn’t downloaded. Larimer County · 6 layers · about 340 MB [**Download Now**] [**Work Online Instead**] [**Dismiss**]

Rules:

- List every missing county with size; offer **Download All Missing** when there are several
- Download proceeds in the background with a progress indicator; the user keeps working
- If a county is partially downloaded, say which layers are missing
- **Never block map creation or export.** In Online mode the map renders live. In Offline mode, uncovered ground renders as an obvious empty pattern rather than silently blank, with a note on screen and a warning in the export preview
- Dismissing is temporary — the banner returns if the area changes again

9.4 Online / Offline mode

A clearly labeled toggle on both the Data screen and in the project header:

- **Online** — stream layers live, nothing stored beyond a temporary cache. Best for exploring new areas.
 - **Offline** — downloaded data only, works with no connection. Layers without local data grey out with a tooltip: *“Not downloaded — switch to Online mode or download this layer on the Data screen.”*
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10. Region Config System (architecture requirement)

- `regions/_national.json` — layers from 7.1, always loaded
- `regions/colorado.json`, `regions/wyoming.json` — state layers from 7.2
- `access-rules/` mirrors this structure for classification rules (6.2)
- Each layer config declares: display name, plain-English description, source type (`xyz`, `wmts`, `geojson`, `shapefile`), source URL, default styling, downloadability, attribute fields to strip on import, and access-classification mapping
- Each state config also declares its **county boundary source**, so the county map in 9.1 is config-driven
- **The layer panel and county map must be generated from these configs at runtime.** Adding a state requires new JSON files and **zero code changes**
- Include `regions/TEMPLATE.json` and `access-rules/TEMPLATE.json` with inline comments, plus a `BUILDING.md` section on adding a state

- Handle malformed or unreachable configs gracefully: skip the layer, log it, tell the user which failed and why in plain English
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11. Cartographic Standards for Printed Output

Output must look like a real topographic map, not a screenshot. Every element below is toggled from the Preview tab (Section 8, Tab 4) and styled from its tab settings.

- **Compass rose** — properly drawn, not a plain arrow. Shows true north and magnetic north with **magnetic declination for the map center labeled in degrees**, calculated from a bundled offline model (WMM/IGRF coefficients) so it works with no internet. Label the model year.
- **Contour lines** — **interval stated in the legend** (e.g. "Contour interval 40 feet"). Index contours heavier and elevation-labeled; intermediate contours lighter and unlabeled. Labels along the line, readable, not colliding.
- **Scale bar** — graphic bar in miles and kilometers, plus the ratio in text.
- **Grid** — None / UTM / MGRS / Latitude-Longitude, labeled at page edges, stating datum (WGS84) and grid zone.
- **Legend** — auto-generated from active layers, including **every visible access category with its exact configured color and pattern**, positioned so it never obscures map content.
- **Title block** — editable title, state/county, generation date, **data source date**, scale, sheet reference.
- **Access disclaimer block** — mandatory per 6.3.
- **Neatline** — clean border framing the map, margin content outside it.
- **Multi-sheet elements** — sheet reference ("Sheet B2 — Page 4 of 6"), index inset showing this sheet within the whole area, edge match markers naming adjoining sheets, user-adjustable overlap margin (default 0.5 inch), and an optional overview sheet as page one.

Printed output is **always light-themed** regardless of the app's light/dark setting — a dark-mode map is not printable.

12. Settings

12.1 Light and dark mode

- Three options: **Light · Dark · Match System**, switchable from Settings and from a quick toggle in the app header.
- Both themes must be fully designed, not a filter over one another — proper contrast, readable at both ends, no washed-out greys.
- The theme applies to app chrome only. Map rendering and all printed output remain light-themed.
- Theme choice persists between sessions and applies instantly with no restart.

12.2 Structure — global settings mirror the tabs

Global Settings has a left sidebar, and **it contains one section for each project tab**, in the same order, so the two are learnable as one system.

Tab-mirrored sections (these set the *defaults for new projects*):

1. **Format** — default paper, orientation, margins, sheet layout, scale, lock behavior, custom paper presets
2. **Layers** — default layers on, default opacities, label sizes, default access color preset
3. **Area** — dim opacity outside the box, margin band style, division line style, sheet labels on map, overlap band visibility, snap-to-north default, coordinate format
4. **Preview** — compass rose style, scale bar units, grid color and weight, legend position, title block contents, contour label size, neatline weight, which elements start checked
5. **Export** — default DPI, output folder, filename pattern, single vs per-sheet, auto-open folder
6. **Notes** — print notes on overview by default, printed note font size

Application-wide sections (not tab-specific): 7. **General** — light/dark/system theme, units (imperial/metric), startup screen, settings difficulty mode 8. **Land Access Colors** — full per-category styling, see 12.4 9. **Projects** — autosave snapshot retention, default fork naming, thumbnail quality, default project folder 10. **Data & Storage** — storage location, online/offline default, cache limit, currency warning threshold, **Backup** and **Restore** 11. **Advanced** — coordinate system, datum, declination model and year, custom tile endpoints, config file locations, hardware acceleration, logging level

The relationship between per-tab and global settings must be stated in the UI, not left implicit:

- A tab's gear icon opens settings **for that project only**.
- The matching global section sets **defaults for new projects**.
- Every per-tab settings drawer has two buttons: **Save as default** (push these values up to global) and **Reset to default** (pull global values down into this project).
- When a project's setting differs from the global default, mark it with a small "modified" indicator so it's obvious why this project looks different.

12.3 Difficulty modes

A selector at the top of Settings with three modes. Changing modes changes **only which settings are visible** — never their values. It applies to per-tab drawers as well as global settings.

- **Beginner** — roughly 12–15 settings. Only things with obvious visible effects. No coordinate systems, datums, or custom endpoints.
- **Intermediate** — roughly 35–45 settings. Adds contour intervals, grid types, tiling overlap, per-category access styling.
- **Advanced** — everything.

Each option carries a one-line description, e.g. *"Beginner — just the essentials. You can switch to a fuller view any time."*

When a user in Beginner or Intermediate searches for a hidden setting, show it in results noting *"This setting is available in Advanced mode"* with a one-click switch.

12.4 Land Access Colors

For **each** of the five categories plus the access-routes line layer:

- **Fill color** — full picker with hex entry
- **Fill opacity** — 0–100% slider
- **Pattern** — Solid / Diagonal hatch / Cross hatch / Dots / None (*patterns survive black-and-white printing and help colorblind users*)
- **Border color** — separate picker
- **Border weight** — in points
- **Border style** — Solid / Dashed / Dotted
- **Label on map** — on/off, with user-editable label text (e.g. rename "Open — Hunting

Allowed" to "Huntable")

- **Show in legend** — on/off, except Unknown, forced on per 6.3

Include a **live preview swatch** per category and a **sample map preview** showing all categories together.

Presets, one click each: *Default · Colorblind Safe · Black & White Print · High Contrast · Subtle*. Users can save named presets and set one as default.

12.5 Settings behavior

- **Every setting has a one-sentence plain-English description under its label.** No exceptions, no jargon-only labels.
 - **Reset to default** per section, plus a global reset with confirmation
 - **Export / Import settings** as JSON for moving between computers
 - Changes apply live where possible; where a restart is needed, say so inline
 - Settings persist automatically — no Save button to hunt for
 - Search box at the top of Settings covering every section
-

13. Explicitly Out of Scope — Do Not Build These

- Weather, forecasts, wind, sunrise/sunset, moon phase, or **any** time-sensitive or live data
- Waypoints, pins, markers, trail cameras, or map annotation (the Notes tab is plain text only)
- GPS tracking, route recording, turn-by-turn navigation, or any in-field function
- **User accounts, logins, cloud sync, servers, telemetry, or analytics of any kind**
- Multi-user or real-time collaboration (project sharing is file-based only, per 5.3)
- Property owner names or any ownership attribute data (7.3)
- **Any claim that the map establishes legal hunting authority** (Section 6)
- Any paid API, metered tile service, or feature requiring an API key
- Mobile builds (iOS/Android)
- Anything requiring the end user to use a command line

14. Build Order

Complete each stage, verify it works, and check in before starting the next.

1. **Skeleton** — Tauri + React + TypeScript. MapLibre rendering the USGS topo basemap. Confirm it builds and runs on Windows, macOS, and Ubuntu. Produce the three installers.
2. **Storage layout** — Section 4 directory structure, SQLite catalog, settings persistence.
3. **Theming** — light and dark modes, system matching, header toggle, applied app-wide.
4. **Projects screen** — create, open, rename, delete, autosave, thumbnails, project cards.
5. **Tabbed workspace shell** — the six tabs in workflow order, navigable, state-preserving, each with its gear icon wired to an (initially empty) settings drawer.
6. **Format tab** — paper, orientation, margins, sheet-layout option cards, scale with lock behavior, freeform toggle.
7. **Layers tab** — `regions/_national.json`, panel generated from it, checkboxes and opacity against live streamed layers.
8. **Area tab — the box** — three-zone shading (bright inside, shaded margin band, dimmed outside), neatline, draggable handles, division lines with overlap band and sheet labels, live sheet/scale readout.
9. **Area tab — the four input methods** — format box, freeform draw, four-corner placement, coordinate entry, all two-way synced.
10. **Access classification engine** — rules files, five categories, Unknown-by-default, source trace, click-to-inspect popup, mandatory warnings, first-run modal.
11. **Single-sheet PDF export** — correct scale and area, active layers, disclaimer block present.
12. **Cartographic elements** — compass rose with offline declination, scale bar, grid, legend, title block, neatline, labeled contours.
13. **Multi-sheet tiling** — grid calculation, overlap, sheet references, index inset, edge match markers, overview sheet.
14. **Preview tab** — paginated preview matching output exactly, with live element checkboxes.
15. **Export tab** — output options, DPI, filename patterns, progress with cancel, export

history.

16. **Settings** — global sections mirroring the tabs, per-tab drawers, Save as default / Reset to default, modified indicators, difficulty modes, search, descriptions on every setting.
 17. **Land access color customization** — full styling controls, live previews, built-in and user presets.
 18. **Region configs** — Colorado and Wyoming layer and rules files, state selector, templates, docs.
 19. **Data screen** — interactive county map with status colors, synced list, layer selection, download manager with resume, Manage Downloads, currency flags, disk usage.
 20. **Missing-county prompt** — coverage detection, non-blocking banner, background download, offline gap rendering.
 21. **Forking and portability** — fork with lineage, snapshot history, project file export/import, backup/restore.
 22. **Polish** — plain-English error handling, tooltips, empty states, app icon, “What this map can’t tell you” help page.
 23. **Packaging & docs** — final installers for all three platforms, user `README.md` , owner `BUILDING.md` .
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15. Definition of Done

The app is finished when a person who has never used GIS software can:

- Install it by double-clicking one file for their operating system
- Download county data without creating a map first, by clicking counties on a map of their state
- Start a new project, pick a paper size, say “two maps,” and see a correctly proportioned box on the map
- See at a glance exactly what will and won’t print, because everything outside the box is dimmed and the margin band is marked
- See where the page split will fall before exporting
- Tick checkboxes in the preview for grid lines and coordinates and watch the map update
- See public-huntable, public-non-hunting, closed, private, and unclassified land in colors

they can change

- Understand, without being told, that those colors are estimates and not legal permission
- Fork a working project to try a variation without risking the original
- Switch between light and dark mode
- Export a professional, correctly scaled, multi-sheet topographic PDF
- Do all of the above with no account, no internet after downloading data, and without ever opening a terminal or editing a file